

CS471 – Web Technologies (Laboratory)		Lab 1 The Internet Protocols
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This lab session covers the usage of the Wireshark application to monitor and capture the outgoing and incoming packets from a network connection (WIFI, ethernet, etc.). Specifically, students should be able to analyze HTTP, HTTPS, TCP/IP, and UDP protocols using Wireshark, a network protocol analyzer, and draw conclusions.

Pre-lab Preparation:

1. Review the basics and the structure of HTTP, TCP/IP, and UDP protocols,
2. Install Wireshark and ensure it is running on your computer,
3. Create an online, *publically accessible* Git repository to host and upload your work in the labs. We recommend you use GitHub or GitLab.

Lab Activities:

Part 1: Capturing HTTP Traffic.

Task 1: Start Wireshark and capture packets.

- Step 1: Open Wireshark.
- Step 2: Select the network interface connected to the internet (e.g., Ethernet or Wi-Fi).
- Step 3: Click the "Start Capturing Packets" button (the shark fin icon).
- Step 4: Open your favorite web browser and navigate to (<http://neverssl.com/>) website.
- Step 5: After the website has fully loaded, stop capturing packets by clicking the red stop button in Wireshark.

Task 2: Filter HTTP packets and analyze them.

- Step 1: In the filter bar, type http and press Enter. This filters out only the HTTP packets from the capture.
- Step 2: Select any HTTP packet to view its details.
- Step 3: Observe the HTTP request and response messages. Note the method (GET, POST), URL, response codes (200 OK, 404 Not Found), etc.

Part 2: Analyzing TCP/IP Traffic.

Task 1: Filter TCP packets

- Step 1:** Clear the previous filter and type TCP to focus on TCP packets.
- Step 2:** Select a TCP packet related to your HTTP request/response.
- Step 3:** Right-click on the packet and select "Follow" -> "TCP Stream".
- Step 4:** This shows the entire conversation between the client and server.

Task 2: Analyze TCP handshake and investigate Data Transfer and Termination

- Step 1:** Find and select packets related to the TCP three-way handshake:
- o SYN: Initiates a connection.
 - o SYN-ACK: Acknowledges and responds to the SYN.
 - o ACK: Acknowledges the SYN-ACK and establishes the connection.
- Step 2:** Note the sequence and acknowledgment numbers. Screenshot and upload your image to your online git repository.
- Step 3:** Observe the data packets exchanged between the client and server. Take a screenshot and upload it to your online git repo.
- Step 4:** Look at the TCP termination process (FIN, ACK packets).

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Part 3: Capturing and Analyzing UDP Traffic

Task 1: Generate UDP traffic and capture packets

Step 1: Open a network application that uses UDP (e.g., streaming video, VoIP software, or custom script).

Step 2: Start the application to generate UDP traffic.

Step 3: Start capturing packets in Wireshark while the UDP application is running.

Step 4: After sufficient traffic is generated, stop capturing packets.

Task 2: Filter and analysis UDP Packets

Step 1: In the filter bar, type UDP and press Enter.

Step 2: This filters out only the UDP packets from the capture.

Step 3: Select any UDP packet to view its details.

Step 4: Observe the source and destination ports, length, and data.

Step 5: Compare the simplicity of UDP headers with TCP headers.

Part 4: Comparing TCP and UDP by filling in the following tables. Save your work (e.g., in an MS Word document), and upload it to your online git repo.

Task 1: Fill in the following table and provide reasons.

	TCP or UDP	Reasons
Reliability and Connection Establishment	TCP	Connection-oriented (uses 3-way handshake) and retransmits lost packets
Data Integrity and Ordering	TCP	Uses sequence numbers to guarantee correct packet order and error checking.

Task 2: Identify the use Cases and Performance of TCP and UDP.

	TCP	UDP
Use cases	Web browsing (HTTP/HTTPS), Email, File transfer.	Video streaming, Online gaming, VoIP, DNS.
Performance	Slower (overhead from handshake and verification).	Faster and lightweight (no connection overhead).